

ADVANCED

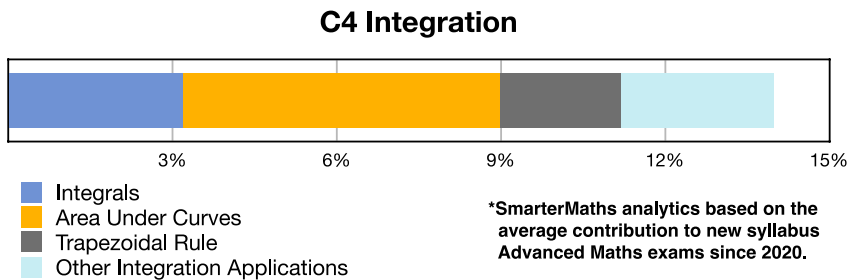
Stage 6

Calculus (Adv), C4 Integration (Adv)

Areas Under Curves (Y12)

Teacher: SHS Maths

Exam Equivalent Time: 174 minutes (based on allocation of 1.5 minutes per mark)



HISTORICAL CONTRIBUTION

- C4 Integration* has contributed a substantial 14.0% to new syllabus Advanced exams since it was introduced in 2020.
- This topic has been split into four sub-topics for analysis purposes: 1-*Integrals* (3.2%), 2-*Areas Under Curves* (5.8%), 3-*Trapezoidal Rule* (2.2%) and 4-*Other Integration Examples* (2.8%).
- This analysis looks at *Areas Under Curves*.

HSC ANALYSIS - What to expect and common pitfalls

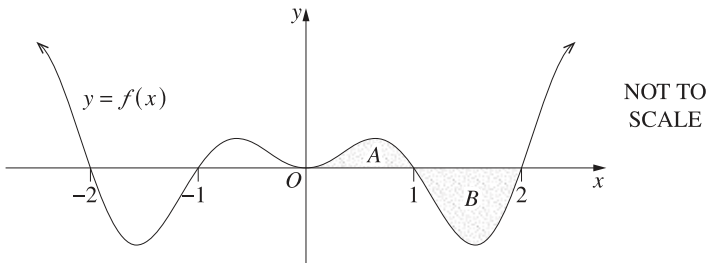
- Areas Under Curves* (5.8%) has attracted an average of two questions per new syllabus exam, appearing in both multiple choice and longer answer sections.
- This topic area can be examined using a very broad range of underlying equations, including quadratic, cubic, exponential, hyperbola and trig functions.
- The most common question type requires students to find areas between given curves and straight lines (although the area between 2 quadratics was asked in 2024 and 2020).
- Important revision questions: 2021 Adv 28 combined Area Under Curves with transformations and 2022 Adv 28 looked at circle functions in this context → both caused problems.
- Definite integrals defining areas for functions that cannot be integrated has been asked via multiple choice 3 times in new syllabus exams, all producing sub-50% mean marks.

Questions

1. Calculus, 2ADV C4 2022 HSC 8 MC

The graph of the even function $y = f(x)$ is shown.

The area of the shaded region A is $\frac{1}{2}$ and the area of the shaded region B is $\frac{3}{2}$.

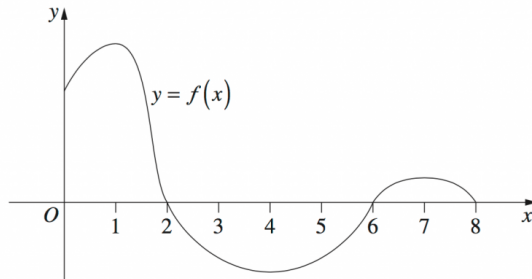


What is the value of $\int_{-2}^2 f(x) \, dx$?

- A. 4
- B. 2
- C. -2
- D. -4

2. Calculus, 2ADV C4 2012 HSC 10 MC

The graph of $y = f(x)$ has been drawn to scale for $0 \leq x \leq 8$.

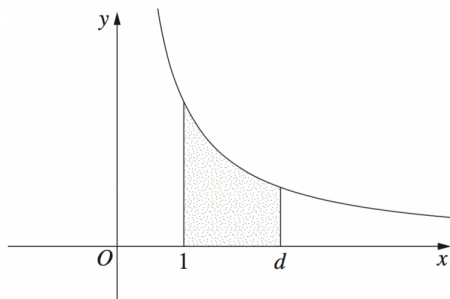


Which of the following integrals has the greatest value?

- A. $\int_0^1 f(x) \, dx$
- B. $\int_0^2 f(x) \, dx$
- C. $\int_0^7 f(x) \, dx$
- D. $\int_0^8 f(x) \, dx$

3. Calculus, 2ADV C4 2015 HSC 10 MC

The diagram shows the area under the curve $y = \frac{2}{x}$ from $x = 1$ to $x = d$.

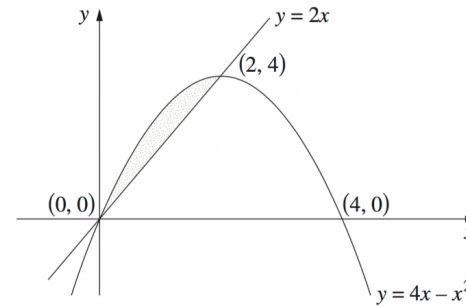


What value of d makes the shaded area equal to 2?

- A. e
- B. $e + 1$
- C. $2e$
- D. e^2

4. Calculus, 2ADV C4 2015 HSC 7 MC

The diagram shows the parabola $y = 4x - x^2$ meeting the line $y = 2x$ at $(0, 0)$ and $(2, 4)$.

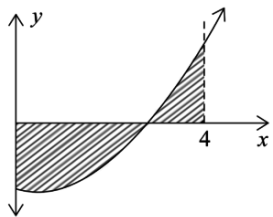


Which expression gives the area of the shaded region bounded by the parabola and the line?

- A. $\int_0^2 x^2 - 2x \, dx$
- B. $\int_0^2 2x - x^2 \, dx$
- C. $\int_0^4 x^2 - 2x \, dx$
- D. $\int_0^4 2x - x^2 \, dx$

5. Calculus, 2ADV C4 SM-Bank 1 MC

The diagram below is the graph of $y = x^2 - x - 6$



What is the correct expression for the area bounded by the x -axis and the graph $y = x^2 - x - 6$ between $0 \leq x \leq 4$?

- A. $A = \int_0^4 x^2 - x - 6 \, dx$
- B. $A = \left| \int_0^3 x^2 - x - 6 \, dx \right| + \int_3^4 x^2 - x - 6 \, dx$
- C. $A = \int_0^3 x^2 - x - 6 \, dx + \left| \int_3^4 x^2 - x - 6 \, dx \right|$
- D. $A = \left| \int_0^4 x^2 - x - 6 \, dx \right|$

6. Calculus, 2ADV C4 2018 HSC 10 MC

A trigonometric function $f(x)$ satisfies the condition

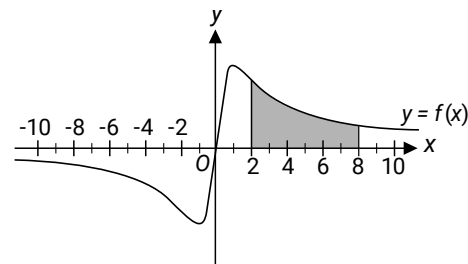
$$\int_0^\pi f(x) \, dx \neq \int_\pi^{2\pi} f(x) \, dx.$$

Which function could be $f(x)$?

- A. $f(x) = \sin(2x)$
- B. $f(x) = \cos(2x)$
- C. $f(x) = \sin\left(\frac{x}{2}\right)$
- D. $f(x) = \cos\left(\frac{x}{2}\right)$

7. Calculus, 2ADV C4 2019 MET-N 15 MC

The area bounded by the graph of $y = f(x)$, the line $x = 2$, the line $x = 8$ and the x -axis, as shaded in the diagram below, is $3\log_e(13)$

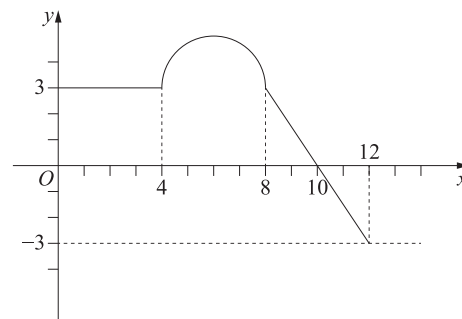


The value of $\int_4^{10} 3f(x-2) \, dx$ is

- A. $3\log_e(13)$
- B. $9\log_e(13)$
- C. $6\log_e(39)$
- D. $9\log_e(11)$

8. Calculus, 2ADV C4 2020 HSC 7 MC

The diagram show the graph $y = f(x)$, which is made up of line segments and a semicircle.



What is the value of $\int_0^{12} f(x) \, dx$?

- A. $24 + 2\pi$
- B. $24 + 4\pi$
- C. $30 + 2\pi$
- D. $30 + 4\pi$

9. Calculus, 2ADV C4 2023 MET2 6 MC

Suppose that $\int_3^{10} f(x) dx = C$ and $\int_7^{10} f(x) dx = D$. The value of $\int_7^3 f(x) dx$ is

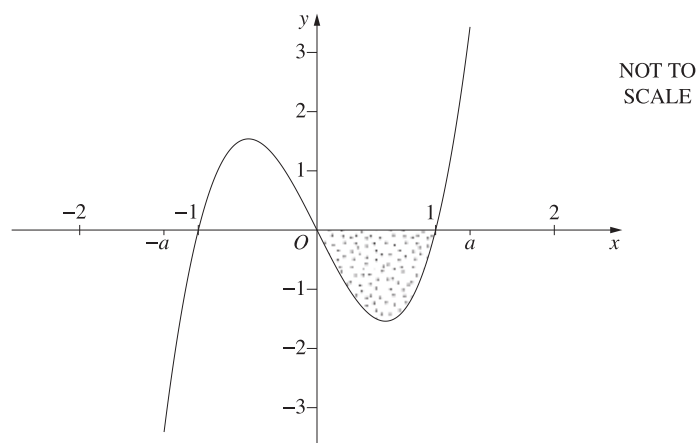
- A. $C + D$
- B. $C + D - 3$
- C. $C - D$
- D. $D - C$

10. Calculus, 2ADV C4 2023 HSC 5 MC

The diagram shows the graph $y = f(x)$, where $f(x)$ is an odd function.

The shaded area is 1 square unit.

The number a , where $a > 1$, is chosen so that $\int_0^a f(x) dx = 0$.

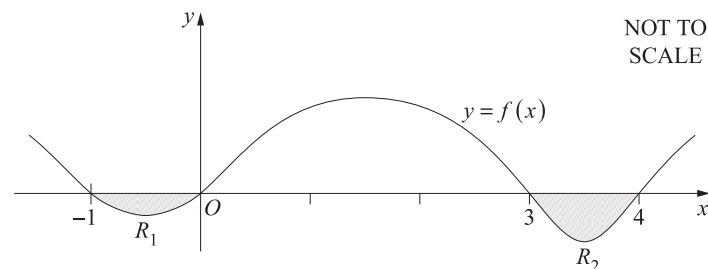


What is the value of $\int_{-a}^1 f(x) dx$?

- A. -1
- B. 0
- C. 1
- D. 3

11. Calculus, 2ADV C4 2018 HSC 7 MC

The diagram shows the graph of $y = f(x)$ with intercepts at $x = -1, 0, 3$ and 4 .



The area of shaded region R_1 is 2.

The area of shaded region R_2 is 3.

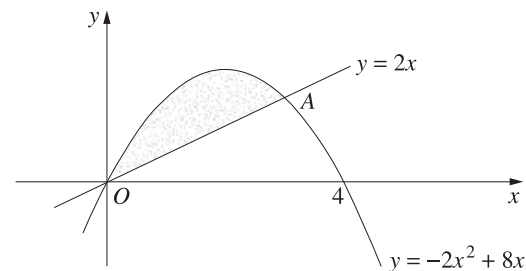
It is given that $\int_0^4 f(x) dx = 10$.

What is the value of $\int_{-1}^3 f(x) dx$?

- A. 5
- B. 9
- C. 11
- D. 15

12. Calculus, 2ADV C4 2014 HSC 12d

The parabola $y = -2x^2 + 8x$ and the line $y = 2x$ intersect at the origin and at the point A .

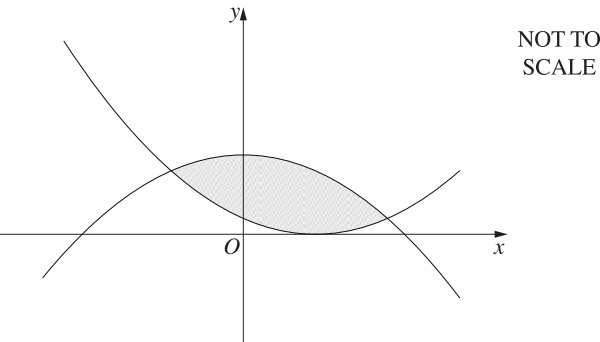


i. Find the x -coordinate of the point A . (1 mark)

ii. Calculate the area enclosed by the parabola and the line. (3 marks)

13. Calculus, 2ADV C4 2024 HSC 14

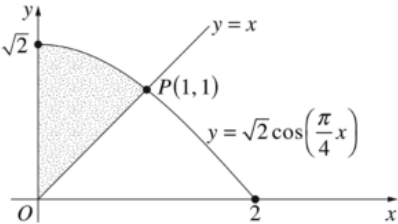
The curves $y = (x - 1)^2$ and $y = 5 - x^2$ intersect at two points, as shown in the diagram.



- a. Find the x -coordinates of the points of intersection of the two curves. (1 mark)
- b. Find the area enclosed by the two curves. (3 marks)

14. Calculus, 2ADV C4 2016 HSC 13d

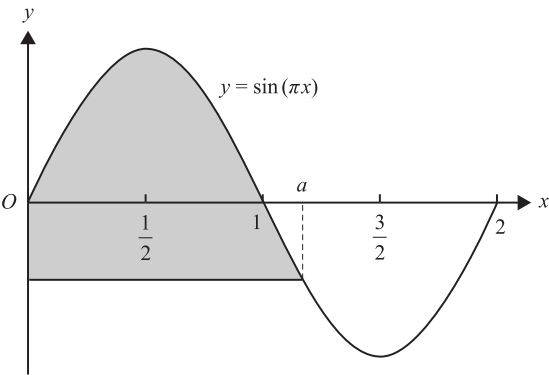
The curve $y = \sqrt{2}\cos\left(\frac{\pi}{4}x\right)$ meets the line $y = x$ at $P(1, 1)$, as shown in the diagram.



Find the exact value of the shaded area. (3 marks)

15. Calculus, 2ADV C4 2019 MET1 7

The shaded region in the diagram below is bounded by the vertical axis, the graph of the function with rule $f(x) = \sin(\pi x)$ and the horizontal line segment that meets the graph at $x = a$, where $1 \leq a \leq \frac{3}{2}$.



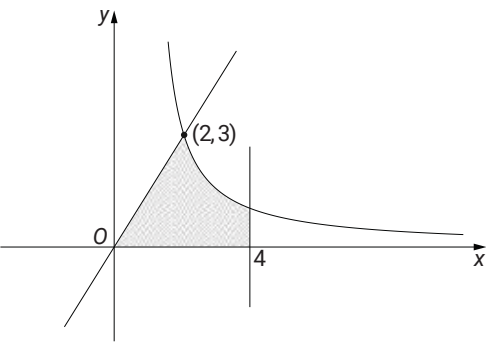
Let $A(a)$ be the area of the shaded region.

Show that $A(a) = \frac{1}{\pi} - \frac{1}{\pi}\cos(a\pi) - a\sin(a\pi)$. (3 marks)

16. Calculus, 2ADV C4 2021 HSC 24

The curve $y = \frac{3}{x - 1}$ intersects the line $y = \frac{3}{2}x$ at the point $(2, 3)$.

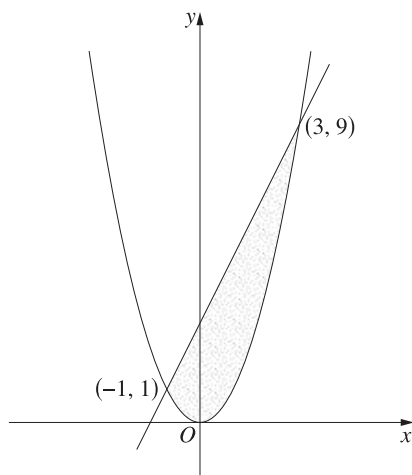
The region bounded by the curve $y = \frac{3}{x - 1}$, the line $y = \frac{3}{2}x$, the x -axis and the line $x = 4$ is shaded in the diagram.



Find the exact area of the shaded region. (3 marks)

17. Calculus, 2ADV C4 2022 HSC 16

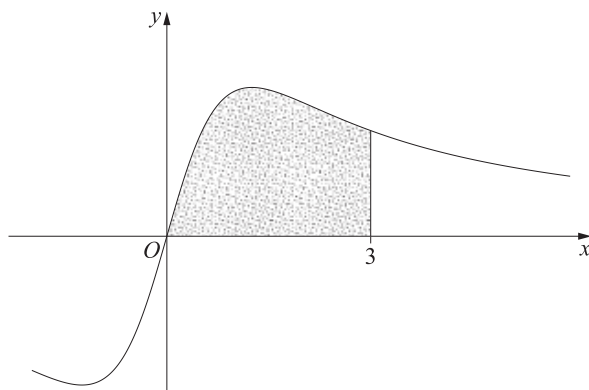
The parabola $y = x^2$ meets the line $y = 2x + 3$ at the points $(-1, 1)$ and $(3, 9)$ as shown in the diagram.



Find the area enclosed by the parabola and the line. **(3 marks)**

18. Calculus, 2ADV C4 2019 HSC 12d

The diagram shows the graph of $y = \frac{3x}{x^2 + 1}$.

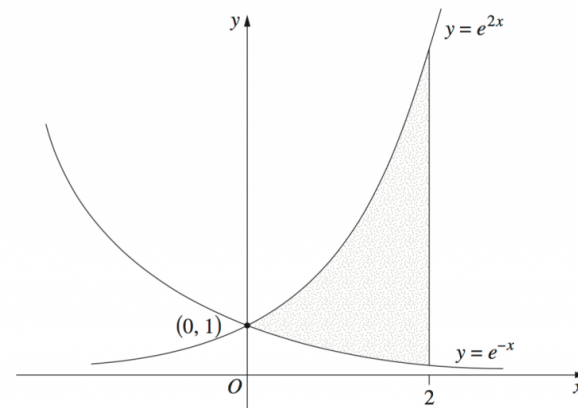


The region enclosed by the graph, the x -axis and the line $x = 3$ is shaded.

Calculate the exact value of the area of the shaded region. **(3 marks)**

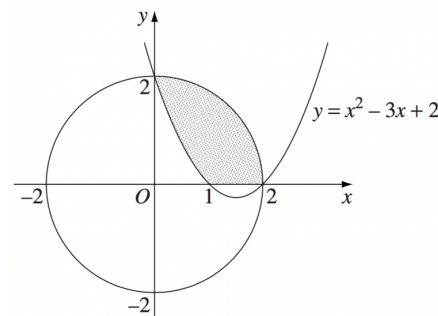
19. Calculus, 2ADV C4 2010 HSC 4b

The curves $y = e^{2x}$ and $y = e^{-x}$ intersect at the point $(0, 1)$ as shown in the diagram.



Find the exact area enclosed by the curves and the line $x = 2$. **(3 marks)**

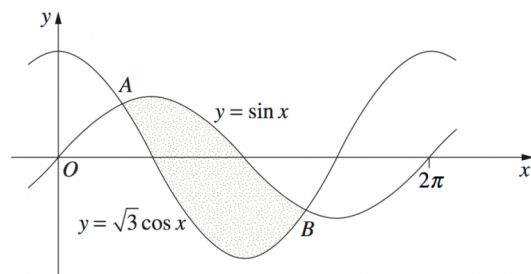
20. Calculus, 2ADV C4 2005 HSC 8b



The shaded region in the diagram is bounded by the circle of radius 2 centred at the origin, the parabola $y = x^2 - 3x + 2$, and the x -axis.

By considering the difference of two areas, find the area of the shaded region. **(3 marks)**

21. Calculus, 2ADV C4 2007 HSC 7b

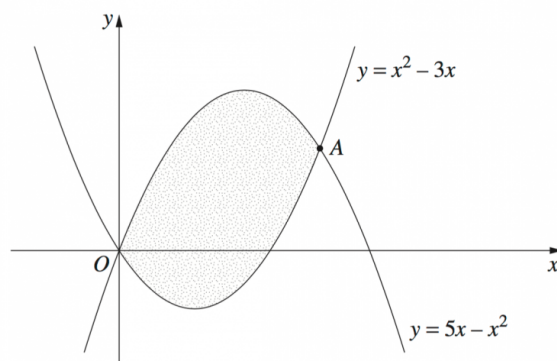


The diagram shows the graphs of $y = \sqrt{3}\cos x$ and $y = \sin x$. The first two points of intersection to the right of the y -axis are labelled A and B .

- Solve the equation $\sqrt{3}\cos x = \sin x$ to find the x -coordinates of A and B . (2 marks)
- Find the area of the shaded region in the diagram. (3 marks)

22. Calculus, 2ADV C4 2012 HSC 13b

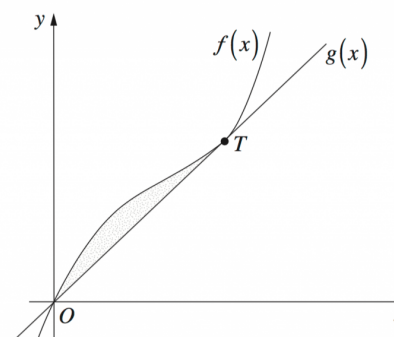
The diagram shows the parabolas $y = 5x - x^2$ and $y = x^2 - 3x$. The parabolas intersect at the origin O and the point A . The region between the two parabolas is shaded.



- Find the x -coordinate of the point A . (1 mark)
- Find the area of the shaded region. (3 marks)

23. Calculus, 2ADV C4 2013 HSC 13b

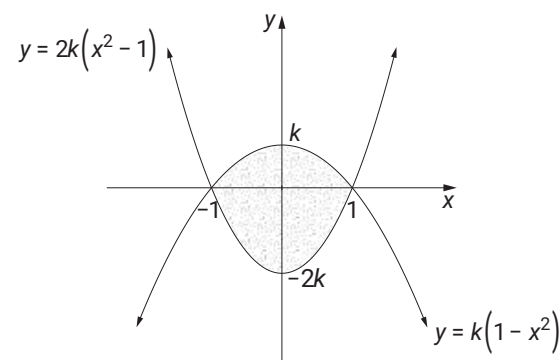
The diagram shows the graphs of the functions $f(x) = 4x^3 - 4x^2 + 3x$ and $g(x) = 2x$. The graphs meet at O and at T .



- Find the x -coordinate of T . (1 mark)
- Find the area of the shaded region between the graphs of the functions $f(x)$ and $g(x)$. (3 marks)

24. Calculus, 2ADV C4 2017 HSC 14d

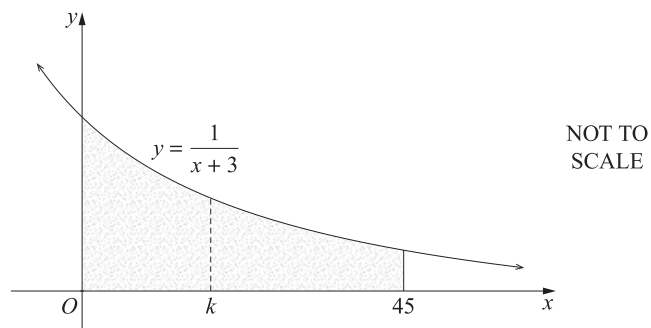
The shaded region shown is enclosed by two parabolas, each with x -intercepts at $x = -1$ and $x = 1$. The parabolas have equations $y = 2k(x^2 - 1)$ and $y = k(1 - x^2)$, where $k > 0$.



Given that the area of the shaded region is 8, find the value of k . (3 marks)

25. Calculus, 2ADV C4 2018 HSC 15b

The diagram shows the region bounded by the curve $y = \frac{1}{x+3}$ and the lines $x = 0$, $x = 45$ and $y = 0$. The region is divided into two parts of equal area by the line $x = k$, where k is a positive integer.



What is the value of the integer k , given that the two parts have equal areas? **(3 marks)**

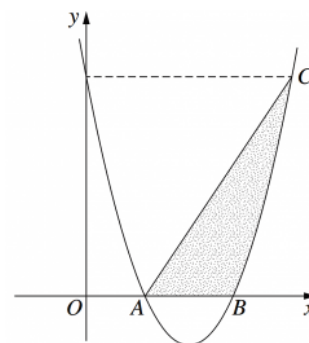
26. Trigonometry, 2ADV T3 2006 HSC 7b

A function $f(x)$ is defined by $f(x) = 1 + 2\cos x$.

- Show that the graph of $y = f(x)$ cuts the x -axis at $x = \frac{2\pi}{3}$. **(1 mark)**
- Sketch the graph of $y = f(x)$ for $-\pi \leq x \leq \pi$ showing where the graph cuts each of the axes. **(3 marks)**
- Find the area under the curve $y = f(x)$ between $x = -\frac{\pi}{2}$ and $x = \frac{2\pi}{3}$. **(3 marks)**

27. Calculus, 2ADV C4 2015 HSC 16a

The diagram shows the curve with equation $y = x^2 - 7x + 10$. The curve intersects the x -axis at points A and B . The point C on the curve has the same y -coordinate as the y -intercept of the curve.

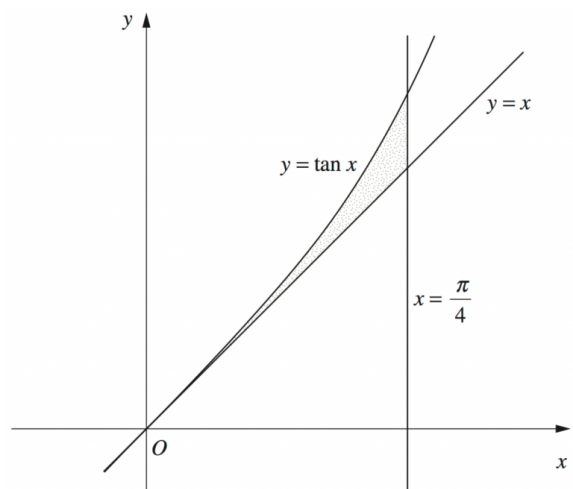


- Find the x -coordinates of points A and B . **(1 mark)**
- Write down the coordinates of C . **(1 mark)**
- Evaluate $\int_0^2 (x^2 - 7x + 10) dx$. **(1 mark)**
- Hence, or otherwise, find the area of the shaded region. **(2 marks)**

28. Calculus, 2ADV C4 2006 HSC 5b

i. Show that $\frac{d}{dx} \log_e(\cos x) = -\tan x$. (1 mark)

ii.

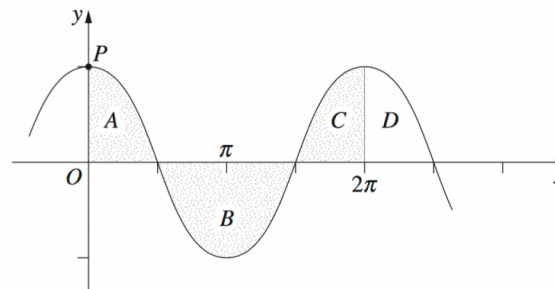


The shaded region in the diagram is bounded by the curve $y = \tan x$ and the lines $y = x$ and $x = \frac{\pi}{4}$.

Using the result of part (i), or otherwise, find the area of the shaded region. (3 marks)

29. Calculus, 2ADV C4 2011 HSC 6c

The diagram shows the graph $y = 2\cos x$.



i. State the coordinates of P . (1 mark)

ii. Evaluate the integral $\int_0^{\frac{\pi}{2}} 2\cos x \, dx$. (2 marks)

iii. Indicate which area in the diagram, A , B , C or D , is represented by the integral

$$\int_{\frac{3\pi}{2}}^{2\pi} 2\cos x \, dx. \quad (1 \text{ mark})$$

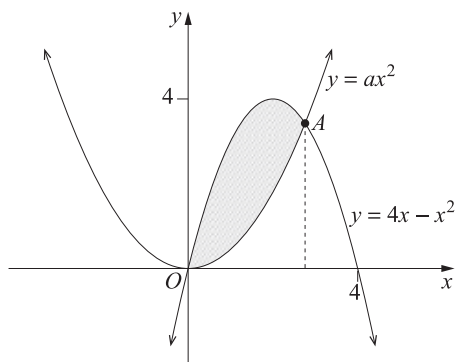
iv. Using parts (ii) and (iii), or otherwise, find the area of the region bounded by the curve $y = 2\cos x$ and the x -axis, between $x = 0$ and $x = 2\pi$. (1 mark)

v. Using the parts above, write down the value of

$$\int_{\frac{\pi}{2}}^{2\pi} 2\cos x \, dx. \quad (1 \text{ mark})$$

30. Calculus, 2ADV C4 2020 HSC 30

The diagram shows two parabolas $y = 4x - x^2$ and $y = ax^2$, where $a > 0$. The two parabolas intersect at the origin, O , and at A .

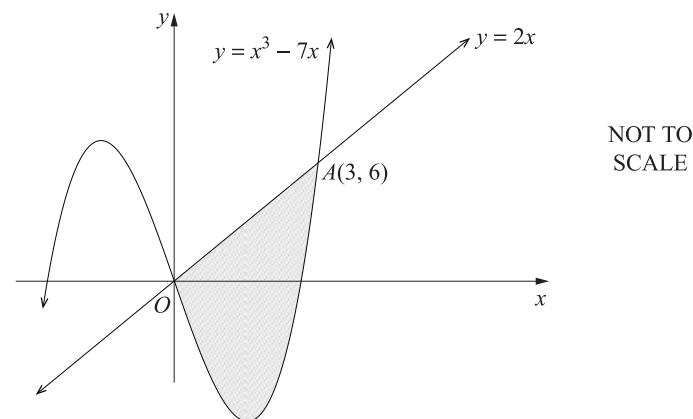


a. Show that the x -coordinate of A is $\frac{4}{a+1}$. (2 marks)

b. Find the value of a such that the shaded area is $\frac{16}{3}$. (4 marks)

31. Calculus, 2ADV C4 2018* HSC 15c

The shaded region is enclosed by the curve $y = x^3 - 7x$ and the line $y = 2x$, as shown in the diagram. The line $y = 2x$ meets the curve $y = x^3 - 7x$ at $O(0, 0)$ and $A(3, 6)$. Do NOT prove this.



i. Use integration to find the area of the shaded region. (2 marks)

ii. Use the Trapezoidal rule and four function values to approximate the area of the shaded region. (2 marks)

The point P is chosen on the curve $y = x^3 - 7x$ so that the tangent at P is parallel to the line $y = 2x$ and the x -coordinate of P is positive

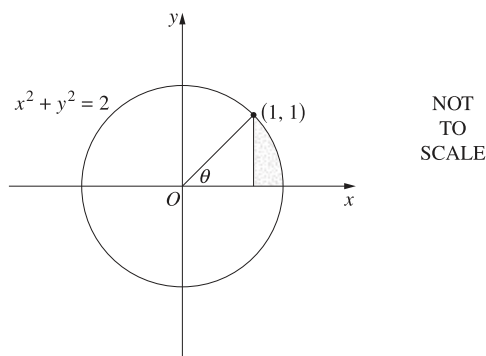
iii. Show that the coordinates of P are $(\sqrt{3}, -4\sqrt{3})$. (2 marks)

iv. Using the perpendicular distance formula $\frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$, find the area of $\triangle OAP$. (2 marks)

32. Calculus, 2ADV C4 2022 HSC 28

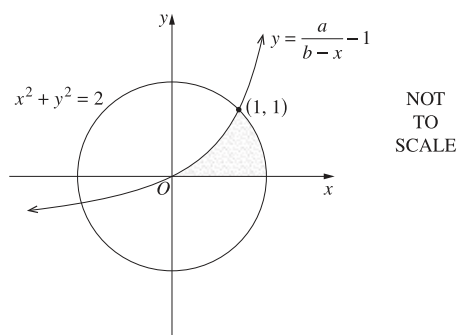
The graph of the circle $x^2 + y^2 = 2$ is shown.

The interval connecting the origin, O , and the point $(1, 1)$ makes an angle θ with the positive x -axis.



a. By considering the value of θ , find the exact area of the shaded region, as shown on the diagram. **(2 marks)**

Part of the hyperbola $y = \frac{a}{b-x} - 1$ which passes through the points $(0, 0)$ and $(1, 1)$ is drawn with the circle $x^2 + y^2 = 2$ as shown.



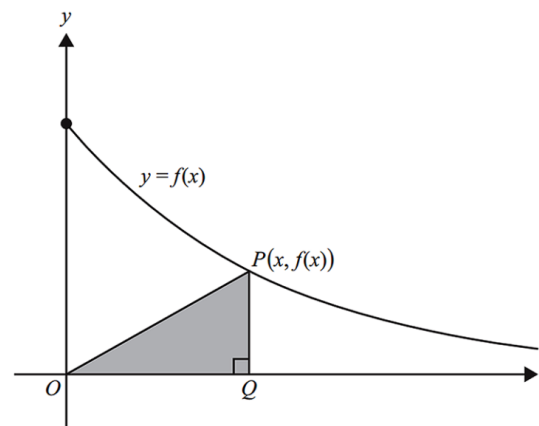
b. Show that $a = b = 2$. **(2 marks)**

c. Using parts (a) and (b), find the exact area of the region bounded by the hyperbola, the positive x -axis and the circle as shown on the diagram. **(3 marks)**

33. Calculus, 2ADV C4 SM-Bank 12

Let $f(x) = 2e^{-\frac{x}{5}}$ for $x \geq 0$

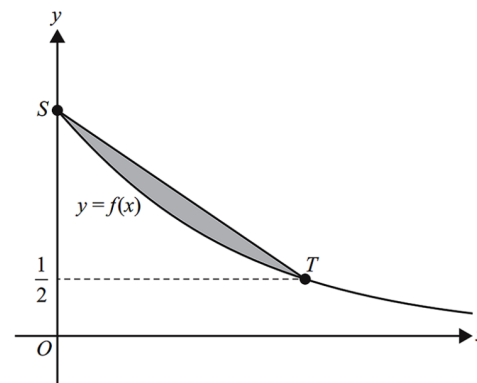
A right-angled triangle OQP has vertex O at the origin, vertex Q on the x -axis and vertex P on the graph of f , as shown. The coordinates of P are $(x, f(x))$.



i. Find the area, A , of the triangle OPQ in terms of x . **(1 mark)**

ii. Find the maximum area of triangle OQP and the value of x for which the maximum occurs. **(3 marks)**

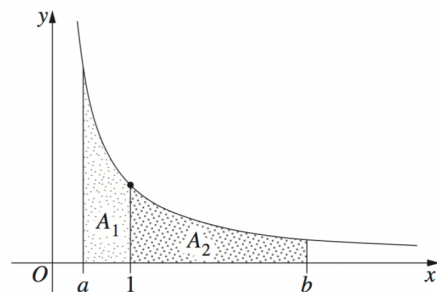
iii. Let S be the point on the graph of f on the y -axis and let T be the point on the graph of f with the y -coordinate $\frac{1}{2}$. Find the area of the region bounded by the graph of f and the line segment ST . **(2 marks)**



34. Calculus, 2ADV C4 2010 HSC 5c

The diagram shows the curve $y = \frac{1}{x}$, for $x > 0$.

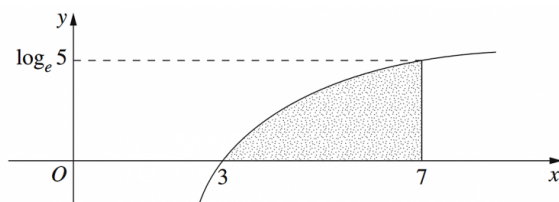
The area under the curve between $x = a$ and $x = 1$ is A_1 . The area under the curve between $x = 1$ and $x = b$ is A_2 .



The areas A_1 and A_2 are each equal to 1 square unit.

Find the values of a and b . (3 marks)

35. Calculus, 2ADV C4 2008 HSC 10a



In the diagram, the shaded region is bounded by $y = \log_e(x-2)$, the x -axis and the line $x = 7$.

Find the exact value of the area of the shaded region. (5 marks)

Worked Solutions

1. Calculus, 2ADV C4 2022 HSC 8 MC

Areas under the x -axis are negative

$$\int_0^2 f(x) \, dx = \frac{1}{2} - \frac{3}{2} = -1$$

Since $f(x)$ is even:

$$\begin{aligned} \int_{-2}^2 f(x) \, dx &= 2 \int_0^2 f(x) \, dx = -2 \\ &\Rightarrow C \end{aligned}$$

Mean mark 52%.

2. Calculus, 2ADV C4 2012 HSC 10 MC

Since the integrals measure the net area under the graph and above the x -axis (i.e. below the x -axis is a negative value.)

$$\Rightarrow B$$

3. Calculus, 2ADV C4 2015 HSC 10 MC

$$\int_1^d \frac{2}{x} \, dx = 2$$

$$[2\log_e x]_1^d = 2$$

$$2\log_e d - 2\log_e 1 = 2$$

$$2\log_e d = 2$$

$$\log_e d = 1$$

$$\therefore d = e$$

$$\Rightarrow A$$

4. Calculus, 2ADV C4 2015 HSC 7 MC

$$\begin{aligned}\text{Shaded Area} &= \int_0^2 4x - x^2 \, dx - \int_0^2 2x \, dx \\ &= \int_0^2 4x - x^2 - 2x \, dx \\ &= \int_0^2 2x - x^2 \, dx\end{aligned}$$

$\Rightarrow B$

5. Calculus, 2ADV C4 SM-Bank 1 MC

$$\begin{aligned}y &= x^2 - x - 6 \\ &= (x-3)(x+2)\end{aligned}$$

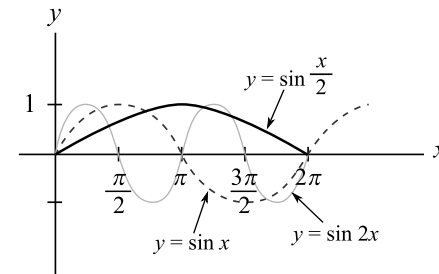
Graph intersects the x -axis at $(3, 0)$

$$\therefore A = \left| \int_0^3 x^2 - x - 6 \, dx \right| + \int_3^4 x^2 - x - 6 \, dx$$

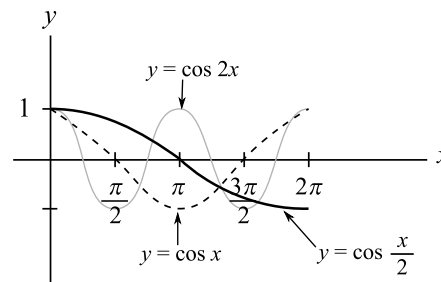
$\Rightarrow B$

6. Calculus, 2ADV C4 2018 HSC 10 MC

Consider options A and C



Consider options B and D



When $y = \cos \frac{x}{2}$,

$$\int_0^\pi f(x) \, dx \neq \int_\pi^{2\pi} f(x) \, dx$$

$\Rightarrow D$

7. Calculus, 2ADV C4 2019 MET-N 15 MC

$$A_1 = \int_2^8 f(x) \, dx = 3 \log_e 13$$

$f(x-2) = f(x)$ shifted 2 units to right.

$$\therefore \int_2^8 f(x) \, dx = \int_4^{10} f(x-2) \, dx$$

$$\begin{aligned}\therefore 3 \int_4^{10} f(x-2) \, dx &= 3 \times 3 \log_e 13 \\ &= 9 \log_e 13\end{aligned}$$

$\Rightarrow B$

8. Calculus, 2ADV C4 2020 HSC 7 MC

Consider the interval between $x = 8$ and $x = 12$:

Area above and below the x -axis are equal.

$$\int_8^{12} f(x) = 0$$

◆ Mean mark 48%.

$$\begin{aligned}\therefore \int_0^{12} f(x) \, dx &= \int_0^8 f(x) \, dx \\ &= \text{Area of rectangle} + \text{area of semi-circle} \\ &= 8 \times 3 + \frac{1}{2} \times \pi \times 2^2 \\ &= 24 + 2\pi\end{aligned}$$

$\Rightarrow A$

9. Calculus, 2ADV C4 2023 MET2 6 MC

Given $\int_3^{10} f(x) \, dx = C$ and $\int_7^{10} f(x) \, dx = D$

We can deduce:

$$\int_3^{10} f(x) \, dx = \int_3^7 f(x) \, dx + \int_7^{10} f(x) \, dx$$

$$C = \int_3^7 f(x) \, dx + D$$

$$C - D = \int_3^7 f(x) \, dx$$

$$\therefore \int_7^3 f(x) \, dx = D - C$$

$\Rightarrow D$

◆ Mean mark 49%.

MARKER'S COMMENT:
41% of students chose option C incorrectly assuming

$$\int_7^3 f(x) \, dx = \int_3^7 f(x) \, dx$$

10. Calculus, 2ADV C4 2023 HSC 5 MC

Since $\int_0^a f(x) \, dx = 0$ and $\int_0^1 f(x) \, dx = -1$ (given)

◆ Mean mark 49%.

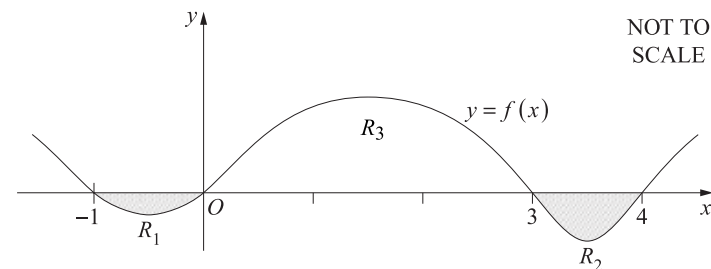
$$\int_1^a f(x) \, dx = 1$$

$$\therefore \int_{-a}^{-1} f(x) \, dx = -1 \quad (f(x) \text{ is odd})$$

$$\therefore \int_{-a}^1 f(x) \, dx = -1$$

$\Rightarrow A$

11. Calculus, 2ADV C4 2018 HSC 7 MC



$$\int_0^4 f(x) \, dx = 10$$

$$\therefore R_3 - R_2 = 10$$

$$R_3 = 13$$

◆ Mean mark 36%.

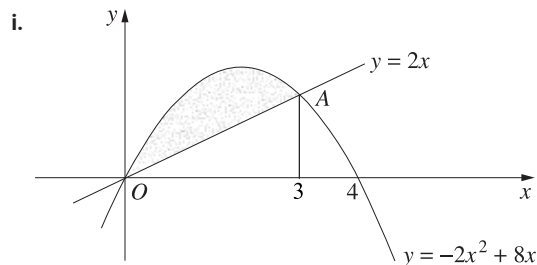
$$\int_{-1}^3 f(x) \, dx = R_3 - R_1$$

$$= 13 - 2$$

$$= 11$$

$\Rightarrow C$

12. Calculus, 2ADV C4 2014 HSC 12d



Need to find x -co-ord of A

$$y = 2x \quad \dots (i)$$

$$y = -2x^2 + 8x \quad \dots (ii)$$

Subst $y = 2x$ from (i) into (ii)

$$-2x^2 + 8x = 2x$$

$$-2x^2 + 6x = 0$$

$$-2x(x - 3) = 0$$

$$\therefore x = 0 \text{ or } 3$$

$$\therefore x\text{-coordinate of } A \text{ is } 3$$

ii.

$$\begin{aligned} \text{Area} &= \int_0^3 (-2x^2 + 8x) \, dx - \int_0^3 2x \, dx \\ &= \int_0^3 (-2x^2 + 6x) \, dx \\ &= \left[-\frac{2}{3}x^3 + 3x^2 \right]_0^3 \\ &= \left[\left(-\frac{2}{3}(3^3) + 3(3^2) \right) - (0 + 0) \right] \\ &= -18 + 27 \\ &= 9 \text{ u}^2 \end{aligned}$$

13. Calculus, 2ADV C4 2024 HSC 14

a. $y = (x - 1)^2$, $y = 5 - x^2$

Intersection occurs when:

$$(x - 1)^2 = 5 - x^2$$

$$x^2 - 2x + 1 = 5 - x^2$$

$$2x^2 - 2x - 4 = 0$$

$$2(x - 2)(x + 1) = 0$$

$$x = 2 \text{ and } -1$$

b.

$$\begin{aligned} \text{Area} &= \int_{-1}^2 (5 - x^2) - (x - 1)^2 \, dx \\ &= \int_{-1}^2 5 - x^2 - x^2 + 2x - 1 \, dx \\ &= \int_{-1}^2 4 - 2x^2 + 2x \, dx \\ &= \left[4x - \frac{2}{3}x^3 + x^2 \right]_{-1}^2 \\ &= \left[\left(8 - \frac{16}{3} + 4 \right) - \left(-4 + \frac{2}{3} + 1 \right) \right] \\ &= \frac{20}{3} - \left(-\frac{7}{3} \right) \\ &= 9 \text{ u}^2 \end{aligned}$$

14. Calculus, 2ADV C4 2016 HSC 13d

Shaded Area

$$\begin{aligned} &= \int_0^1 \sqrt{2} \cos\left(\frac{\pi}{4}x\right) \, dx - \int_0^1 x \, dx \\ &= \int_0^1 \left(\sqrt{2} \cos\left(\frac{\pi}{4}x\right) - x \right) \, dx \\ &= \left[\sqrt{2} \times \frac{4}{\pi} \sin\left(\frac{\pi}{4}x\right) - \frac{x^2}{2} \right]_0^1 \\ &= \left[\left(\frac{4\sqrt{2}}{\pi} \sin \frac{\pi}{4} - \frac{1}{2} \right) - 0 \right] \\ &= \frac{4\sqrt{2}}{\pi} \times \frac{1}{\sqrt{2}} - \frac{1}{2} \\ &= \left(\frac{4}{\pi} - \frac{1}{2} \right) \text{ u}^2 \end{aligned}$$

15. Calculus, 2ADV C4 2019 MET1 7

Lower border of shaded area: $y = f(a) = \sin(a\pi)$

$$\begin{aligned}\text{Area} &= \int_0^a \sin(\pi x) - \sin(a\pi) \, dx \\ &= \left[-\frac{1}{\pi} \cos(\pi x) - x \sin(a\pi) \right]_0^a \\ &= \left[-\frac{1}{\pi} \cos(\pi a) - a \sin(a\pi) - \left(-\frac{1}{\pi} - 0 \right) \right] \\ &= \frac{1}{\pi} - \frac{1}{\pi} \cos(a\pi) - a \sin(a\pi)\end{aligned}$$

16. Calculus, 2ADV C4 2021 HSC 24

$$\begin{aligned}\text{Area} &= \int_0^2 \frac{3}{2}x \, dx + \int_2^4 \frac{3}{x-1} \, dx \\ &= \left[\frac{3}{4}x^2 \right]_0^2 + 3[\log_e(x-1)]_2^4 \\ &= \frac{12}{4} + 3[\log_e 3 - \log_e 1] \\ &= 3 + 3\log_e 3 \quad \text{u}^2\end{aligned}$$

17. Calculus, 2ADV C4 2022 HSC 16

$$\begin{aligned}A &= \int_{-1}^3 2x + 3 - x^2 \, dx \\ &= \left[x^2 + 3x - \frac{1}{3}x^3 \right]_{-1}^3 \\ &= \left(9 + 9 - \frac{27}{3} \right) - \left(1 - 3 + \frac{1}{3} \right) \\ &= 9 - \left(-\frac{5}{3} \right) \\ &= \frac{32}{3} \quad \text{u}^2\end{aligned}$$

18. Calculus, 2ADV C4 2019 HSC 12d

$$\begin{aligned}\text{Area} &= \int_0^3 \frac{3x}{x^2 + 1} \, dx \\ &= \frac{3}{2} \int_0^3 \frac{2x}{x^2 + 1} \, dx \\ &= \frac{3}{2} [\ln(x^2 + 1)]_0^3 \\ &= \frac{3}{2} (\ln 10 - \ln 1) \\ &= \frac{3}{2} \ln 10 \quad \text{u}^2\end{aligned}$$

19. Calculus, 2ADV C4 2010 HSC 4b

$$\begin{aligned}\text{Area} &= \int_0^2 e^{2x} \, dx - \int_0^2 e^{-x} \, dx \\ &= \int_0^2 (e^{2x} - e^{-x}) \, dx \\ &= \left[\frac{1}{2}e^{2x} + e^{-x} \right]_0^2 \\ &= \left[\left(\frac{1}{2}e^4 + e^{-2} \right) - \left(\frac{1}{2}e^0 + e^0 \right) \right] \\ &= \frac{1}{2}e^4 + e^{-2} - \frac{3}{2} \quad \text{u}^2\end{aligned}$$

MARKER'S COMMENT: The best responses used only a *single integral* before any substitution as shown in Worked Solutions.

20. Calculus, 2ADV C4 2005 HSC 8b

Shaded Area = Area in the quarter circle less
the area below the parabola between $x = 0$ and 1.

$$\begin{aligned}\text{Area of } \frac{1}{4} \text{ circle} &= \frac{1}{4} \times \pi r^2 \\ &= \frac{1}{4} \times \pi \times 2^2 \\ &= \pi \text{ u}^2\end{aligned}$$

Area below the parabola between $x = 0$ and 1

$$\begin{aligned}&= \int_0^1 y \, dx \\ &= \int_0^1 x^2 - 3x + 2 \, dx \\ &= \left[\frac{x^3}{3} - \frac{3}{2}x^2 + 2x \right]_0^1 \\ &= \left[\left(\frac{1}{3} - \frac{3}{2} + 2 \right) - 0 \right] \\ &= \frac{5}{6}\end{aligned}$$

$$\therefore \text{Shaded Area} = \left(\pi - \frac{5}{6} \right) \text{ u}^2$$

21. Calculus, 2ADV C4 2007 HSC 7b

i. $\sqrt{3}\cos x = \sin x$

$$\tan x = \sqrt{3}$$

Since $\tan \frac{\pi}{3} = \sqrt{3}$ and $\tan$ is

positive in 1st/3rd quadrants:

$$\begin{aligned}x &= \frac{\pi}{3}, \pi + \frac{\pi}{3} \\ &= \frac{\pi}{3}, \frac{4\pi}{3}\end{aligned}$$

$\therefore$ The x -coordinates of A and B

$$\text{are } x = \frac{\pi}{3}, \frac{4\pi}{3}.$$

ii. Shaded Area

$$\begin{aligned}&= \int_{\frac{\pi}{3}}^{\frac{4\pi}{3}} \sin x \, dx - \int_{\frac{\pi}{3}}^{\frac{4\pi}{3}} \sqrt{3} \cos x \, dx \\ &= \int_{\frac{\pi}{3}}^{\frac{4\pi}{3}} \sin x - \sqrt{3} \cos x \, dx \\ &= \left[-\cos x - \sqrt{3} \sin x \right]_{\frac{\pi}{3}}^{\frac{4\pi}{3}} \\ &= \left[\left(-\cos \frac{4\pi}{3} - \sqrt{3} \sin \frac{4\pi}{3} \right) - \left(-\cos \frac{\pi}{3} - \sqrt{3} \sin \frac{\pi}{3} \right) \right] \\ &= \left[\left(\frac{1}{2} + \sqrt{3} \times \frac{\sqrt{3}}{2} \right) - \left(-\frac{1}{2} - \sqrt{3} \times \frac{\sqrt{3}}{2} \right) \right] \\ &= [2 - (-2)] \\ &= 4 \text{ u}^2\end{aligned}$$

22. Calculus, 2ADV C4 2012 HSC 13b

i. $y = x^2 - 3x \dots (1)$

$y = 5x - x^2 \dots (2)$

Solve: $(1) = (2)$

$x^2 - 3x = 5x - x^2$

$2x^2 - 8x = 0$

$2x(x - 4) = 0$

$x = 0 \text{ or } 4$

$\therefore$ x -coordinate of A is 4

ii. Area $= \int_0^4 (5x - x^2) dx - \int_0^4 (x^2 - 3x) dx$

$= \int_0^4 (5x - x^2 - x^2 + 3x) dx$

$= \int_0^4 (8x - 2x^2) dx$

$= \left[4x^2 - \frac{2}{3}x^3 \right]_0^4$

$= \left[(4 \times 4^2) - \left(\frac{2}{3} \times 4^3 \right) \right]$

$= \left[64 - \frac{128}{3} \right]$

$= \frac{64}{3} \text{ u}^2$

MARKER'S COMMENT: Less errors were made in part (ii) by students who simplified the definite integral before integrating, as shown in the worked solution.

23. Calculus, 2ADV C4 2013 HSC 13b

i. T occurs when $f(x) = g(x)$

$4x^3 - 4x^2 + 3x = 2x$

$4x^3 - 4x^2 + x = 0$

$x(4x^2 - 4x + 1) = 0$

$x(2x - 1)^2 = 0$

$2x - 1 = 0 \Rightarrow x = \frac{1}{2}$

Substitute $x = \frac{1}{2}$ into $g(x)$

$g\left(\frac{1}{2}\right) = 2 \times \frac{1}{2} = 1$

$\therefore T\left(\frac{1}{2}, 1\right)$

ii. Shaded Area $= \int_0^{\frac{1}{2}} (f(x) - g(x)) dx$

$= \int_0^{\frac{1}{2}} (4x^3 - 4x^2 + x) dx$

$= \left[x^4 - \frac{4}{3}x^3 + \frac{1}{2}x^2 \right]_0^{\frac{1}{2}}$

$= \left[\left(\left(\frac{1}{2} \right)^4 - \frac{4}{3} \left(\frac{1}{2} \right)^3 + \frac{1}{2} \left(\frac{1}{2} \right)^2 \right) - 0 \right]$

$= \frac{1}{16} - \frac{1}{6} + \frac{1}{8}$

$= \frac{1}{48} \text{ u}^2$

$\therefore$ Shaded area between the curves is $\frac{1}{48} \text{ u}^2$

24. Calculus, 2ADV C4 2017 HSC 14d

$$\text{Area} = \int_{-1}^1 k(1-x^2) dx - \int_{-1}^1 2k(x^2-1) dx$$

$$8 = 2 \int_0^1 k - kx^2 - 2kx^2 + 2k dx$$

$$8 = 2 \int_0^1 3k - 3kx^2 dx$$

$$8 = 2 [3kx - kx^3]_0^1$$

$$8 = 2[(3k - k) - 0]$$

$$8 = 4k$$

$$\therefore k = 2$$

25. Calculus, 2ADV C4 2018 HSC 15b

$$\text{Total Area} = \int_0^{45} \frac{1}{x+3}$$

$$= [\ln(x+3)]_0^{45}$$

$$= \ln 48 - \ln 3$$

$$= \ln 16$$

$$\Rightarrow \int_0^k \frac{1}{x+3} = \frac{1}{2} \times \ln 16$$

$$[\ln(x+3)]_0^k = \ln 16^{\frac{1}{2}}$$

$$\ln(k+3) - \ln 3 = \ln 4$$

$$\ln\left(\frac{k+3}{3}\right) = \ln 4$$

$$\frac{k+3}{3} = 4$$

$$\therefore k = 4 \times 3 - 3$$

$$= 9$$

26. Trigonometry, 2ADV T3 2006 HSC 7b

i. $f(x) = 1 + 2\cos x$

$$f(x) \text{ cuts the } x\text{-axis when } f(x) = 0$$

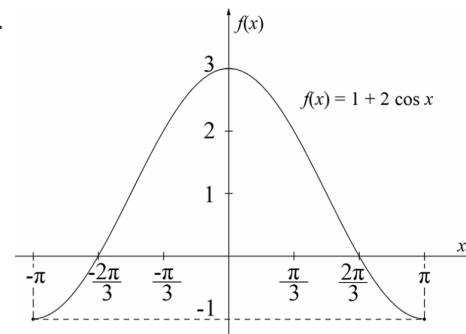
$$1 + 2\cos x = 0$$

$$2\cos x = -1$$

$$\cos x = -\frac{1}{2}$$

$$\therefore x = \frac{2\pi}{3} \dots \text{ as required}$$

ii.



iii. $\text{Area} = \int_{-\frac{\pi}{2}}^{\frac{2\pi}{3}} 1 + 2\cos x dx$

$$= [x + 2\sin x]_{-\frac{\pi}{2}}^{\frac{2\pi}{3}}$$

$$= \left[\left(\frac{2\pi}{3} + 2\sin \frac{2\pi}{3} \right) - \left(-\frac{\pi}{2} + 2\sin \frac{-\pi}{2} \right) \right]$$

$$= \left(\frac{2\pi}{3} + 2 \times \frac{\sqrt{3}}{2} \right) - \left(-\frac{\pi}{2} + 2(-1) \right)$$

$$= \frac{2\pi}{3} + \sqrt{3} + \frac{\pi}{2} + 2$$

$$= \left(\frac{7\pi}{6} + \sqrt{3} + 2 \right) \text{ u}^2$$

27. Calculus, 2ADV C4 2015 HSC 16a

i. $y = x^2 - 7x + 10$
 $= (x - 2)(x - 5)$

$\therefore x = 2$ or 5
 $\therefore$ x -coordinate of $A = 2$
 x -coordinate of $B = 5$

ii. y intercept occurs when $x = 0$
 $\Rightarrow y$ -intercept $= 10$

C occurs at intercept:

$y = x^2 - 7x + 10 \quad \dots (1)$
 $y = 10 \quad \dots (2)$

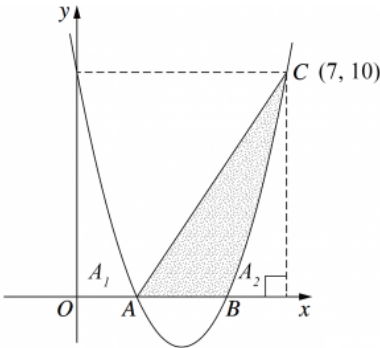
$(1) = (2)$

$x^2 - 7x + 10 = 10$
 $x^2 - 7x = 10$
 $x(x - 7) = 10$

$x = 0$ or 7
 $\therefore C$ is $(7, 10)$

iii. $\int_0^2 (x^2 - 7x + 10) \, dx$
 $= \left[\frac{1}{3}x^3 - \frac{7}{2}x^2 + 10x \right]_0^2$
 $= \left[\left(\frac{1}{3} \times 2^3 - \frac{7}{2} \times 2^2 + 10 \times 2 \right) - 0 \right]$
 $= \frac{8}{3} - 14 + 20$
 $= 8\frac{2}{3}$

iv.



$A_1 = A_2$
 $A_2 = 8\frac{2}{3} \text{ u}^2 \quad (\text{from part (iii)})$

Let D be $(7, 0)$

Shaded Area
 $= \text{Area of } \triangle ACD - A_2$
 $= \frac{1}{2}bh - 8\frac{2}{3}$
 $= \frac{1}{2} \times 5 \times 10 - 8\frac{2}{3}$
 $= 16\frac{1}{3} \text{ u}^2$

♦ Mean mark 49%.

28. Calculus, 2ADV C4 2006 HSC 5b

i. $\frac{d}{dx}\log_e(\cos x) = \frac{-\sin x}{\cos x}$
 $= -\tan x \dots$ as required

ii. Shaded Area

$$\begin{aligned} &= \int_0^{\frac{\pi}{4}} \tan x \, dx - \int_0^{\frac{\pi}{4}} x \, dx \\ &= \int_0^{\frac{\pi}{4}} \tan x - x \, dx \\ &= \left[-\log_e(\cos x) - \frac{1}{2}x^2 \right]_0^{\frac{\pi}{4}} \\ &= \left[\left(-\log_e\left(\cos \frac{\pi}{4}\right) - \frac{1}{2} \times \frac{\pi^2}{16} \right) - \left(-\log_e(\cos 0) - 0 \right) \right] \\ &= -\log_e \frac{1}{\sqrt{2}} - \frac{\pi^2}{32} + \log_e 1 \\ &= -\log_e 2^{-\frac{1}{2}} - \frac{\pi^2}{32} \\ &= \left(\frac{1}{2} \log_e 2 - \frac{\pi^2}{32} \right) \text{ u}^2 \end{aligned}$$

29. Calculus, 2ADV C4 2011 HSC 6c

i. $y = 2\cos x$
At $x = 0$
 $y = 2\cos 0 = 2$
 $\therefore P(0, 2)$

ii. $\int_0^{\frac{\pi}{2}} 2\cos x \, dx$
 $= [2\sin x]_0^{\frac{\pi}{2}}$
 $= \left[2\sin\left(\frac{\pi}{2}\right) - 2\sin 0 \right]$
 $= 2 - 0$
 $= 2$

iii. C

iv. Since Area $A =$ Area C , and
Area $B = 2 \times$ Area A

$$\begin{aligned} \therefore \text{Total Area} &= 2 + (2 \times 2) + 2 \\ &= 8 \text{ u}^2 \end{aligned}$$

v. $\int_{\frac{\pi}{2}}^{2\pi} 2\cos x \, dx$
 $= \text{Area } C - \text{Area } B$
 $= 2 - (2 \times 2)$
 $= -2$

MARKER'S COMMENT: "Using the parts above" in part (v) was ignored by many students. Important to know when finding an area and evaluating an integral may differ.

30. Calculus, 2ADV C4 2020 HSC 30

a. Intersection occurs when

$$\begin{aligned}4x-x^2 &= ax^2 \\ x^2(a+1)-4x &= 0 \\ x[x(a+1)-4] &= 0 \\ x(a+1)-4 &= 0 \quad \text{or} \quad x=0 \\ \therefore x_A &= \frac{4}{a+1}\end{aligned}$$

b. $\text{Area} = \int_0^{\frac{4}{a+1}} 4x-x^2 \, dx - \int_0^{\frac{4}{a+1}} ax^2 \, dx$

$$\begin{aligned}\frac{16}{3} &= \int_0^{\frac{4}{a+1}} 4x-(1+a)x^2 \, dx \\ \frac{16}{3} &= \left[2x^2 - \left(\frac{1+a}{3} \right) x^3 \right]_0^{\frac{4}{a+1}} \\ \frac{16}{3} &= 2\left(\frac{4}{a+1} \right)^2 - \left(\frac{1+a}{3} \right) \left(\frac{4}{a+1} \right)^3 \\ \frac{16}{3} &= \frac{32}{(a+1)^2} - \frac{64}{3} \cdot \frac{1}{(a+1)^2} \\ 16 &= \frac{96}{(a+1)^2} - \frac{64}{(a+1)^2} \\ 16(a+1)^2 &= 32 \\ (a+1)^2 &= 2 \\ a+1 &= \sqrt{2}, \quad (a > 0) \\ \therefore a &= \sqrt{2}-1\end{aligned}$$

♦ Mean mark part (b) 48%.

31. Calculus, 2ADV C4 2018* HSC 15c

i. $\text{Area} = \int_0^3 2x-(x^3-7x) \, dx$

$$\begin{aligned}&= \int_0^3 9x-x^3 \, dx \\ &= \left[\frac{9}{2}x^2 - \frac{1}{4}x^4 \right]_0^3 \\ &= \left[\left(\frac{9}{2} \times 3^2 - \frac{1}{4} \times 3^4 \right) - 0 \right] \\ &= \frac{81}{2} - \frac{81}{4} \\ &= \frac{81}{4} \text{ units}^2\end{aligned}$$

ii. $f(x) = 9x-x^3$

x	0	1	2	3
$f(x)$	0	8	10	0
weight	1	2	2	1

$$\begin{aligned}\text{Area} &\approx \frac{1}{2}[0+2(8+10)+0] \\ &\approx \frac{1}{2}(36) \\ &\approx 18 \text{ u}^2\end{aligned}$$

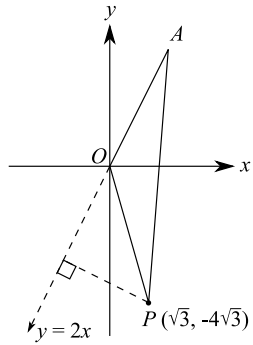
iii. $y = x^3-7x$

$$\begin{aligned}\frac{dy}{dx} &= 3x^2-7 \\ \text{Find } x \text{ such that } \frac{dy}{dx} &= 2 \\ 3x^2-7 &= 2 \\ 3x^2 &= 9 \\ x^2 &= 3 \\ x &= \sqrt{3} \quad (x > 0)\end{aligned}$$

$$\begin{aligned}y &= \left(\sqrt{3} \right)^3 - 7\sqrt{3} \\ &= 3\sqrt{3} - 7\sqrt{3} \\ &= -4\sqrt{3}\end{aligned}$$

$$\therefore P \text{ has coordinates } \left(\sqrt{3}, -4\sqrt{3} \right)$$

iv.



$$\begin{aligned}\text{dist } OA &= \sqrt{(3-0)^2 + (6-0)^2} \\ &= \sqrt{45} \\ &= 3\sqrt{5}\end{aligned}$$

Find $\perp$ distance of P from OA

$$P(\sqrt{3}, -4\sqrt{3}), \quad 2x - y = 0$$

$$\begin{aligned}\perp \text{ dist} &= \left| \frac{2\sqrt{3} + 4\sqrt{3}}{\sqrt{3} + 2} \right| \\ &= \frac{6\sqrt{3}}{\sqrt{5}}\end{aligned}$$

$$\begin{aligned}\therefore \text{Area} &= \frac{1}{2} \times 3\sqrt{5} \times \frac{6\sqrt{3}}{\sqrt{5}} \\ &= 9\sqrt{3} \text{ units}^2\end{aligned}$$

32. Calculus, 2ADV C4 2022 HSC 28

a. $\tan \theta = 1 \Rightarrow \theta = \frac{\pi}{4}$

Using Pythagoras,

$$r = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\text{Shaded Area} = A_{\text{sector}} - A_{\Delta}$$

$$\begin{aligned}&= \frac{\frac{\pi}{4}}{2\pi} \times \pi r^2 - \frac{1}{2} \times b \times h \\ &= \frac{1}{8} \times \pi \times (\sqrt{2})^2 - \frac{1}{2} \times 1 \times 1 \\ &= \frac{\pi}{4} - \frac{1}{2} \\ &= \frac{\pi - 2}{4} \text{ u}^2\end{aligned}$$

Mean mark (a) 54%.

b. Show $a = b = 2$

$$y = \frac{a}{b-x} - 1 \text{ passes through } (0, 0):$$

$$0 = \frac{a}{b-0} - 1$$

$$\frac{a}{b} = 1$$

$$a = b$$

$$y = \frac{a}{b-x} - 1 \text{ passes through } (1, 1):$$

$$1 = \frac{a}{b-1} - 1$$

$$\frac{a}{b-1} = 2$$

$$a = 2(b-1)$$

$$a = 2b - 2$$

$$b = 2b - 2 \text{ (using } a = b)$$

$$b = 2$$

$$\therefore a = b = 2$$

♦ Mean mark (b) 47%.

c. $\int_0^1 \frac{2}{2-x} - 1 \, dx = \int_0^1 -2 \times \frac{-1}{2-x} - 1 \, dx$

$$= [-2\ln|2-x| - x]_0^1$$

$$= [(-2\ln 1 - 1) - (-2\ln 2 - 0)]$$

$$= -1 + 2\ln 2$$

♦♦ Mean mark (c) 36%.

$$\begin{aligned}\therefore \text{ Total Area} &= 2\ln 2 - 1 + \frac{\pi - 2}{4} \\ &= \frac{8\ln 2 - 4 + \pi - 2}{4} \\ &= \frac{8\ln 2 + \pi - 6}{4} \text{ u}^2\end{aligned}$$

33. Calculus, 2ADV C4 SM-Bank 12

$$\begin{aligned}\text{i. Area} &= \frac{1}{2} \times b \times h \\ &= \frac{1}{2}x\left(2e^{-\frac{x}{5}}\right)\end{aligned}$$

$$\therefore A = xe^{-\frac{x}{5}}$$

$$\text{ii. Stationary point when } \frac{dA}{dx} = 0,$$

$$x\left(-\frac{1}{5}e^{-\frac{x}{5}}\right) + e^{-\frac{x}{5}} = 0$$

$$e^{-\frac{x}{5}}\left(1-\frac{x}{5}\right) = 0$$

$$\therefore x = 5 \quad \left(e^{-\frac{x}{5}} > 0, \text{ for all } x\right)$$

$$\text{When } x = 5, \ A = xe^{-\frac{x}{5}} = 5e^{-1}$$

$$\therefore A_{\max} = \frac{5}{e} \text{ u}^2, \text{ when } x = 5$$

$$\begin{aligned}\text{iii. Find } S: F(0) &= 2 \\ \Rightarrow S(0, 2)\end{aligned}$$

$$\text{Find } T: \ 2e^{-\frac{x}{5}} = \frac{1}{2}$$

$$e^{-\frac{x}{5}} = \frac{1}{4}$$

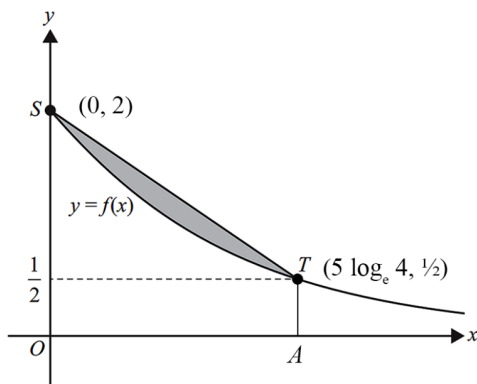
$$-\frac{x}{5} = \log_e\left(\frac{1}{4}\right)$$

$$x = 5\log_e(4)$$

$$\Rightarrow T\left(5\log_e(4), \frac{1}{2}\right)$$

◆ Mean mark (Vic) 35%.

◆◆ Mean mark (Vic) 32%.



$$\begin{aligned}
 \therefore \text{Area} &= \text{Area } SOAT - \int_0^{5\log_e(4)} (2e^{-\frac{x}{5}}) dx \\
 &= \frac{1}{2}h(a+b) + 10 \left[e^{-\frac{x}{5}} \right]_0^{5\log_e(4)} \\
 &= \frac{5}{2} \log_e(4) \left(2 + \frac{1}{2} \right) + 10 \left[e^{-\log_e(4)} - e^0 \right] \\
 &= \frac{25}{4} \log_e(4) + 10 \left(\frac{1}{4} - 1 \right) \\
 &= \frac{25}{4} \log_e(4) - \frac{15}{2} \text{ u}^2
 \end{aligned}$$

34. Calculus, 2ADV C4 2010 HSC 5c

$$\int_a^1 \frac{1}{x} dx = 1$$

$$[\ln x]_a^1 = 1$$

$$\ln 1 - \ln a = 1$$

$$\ln a = -1$$

$$\therefore a = e^{-1} = \frac{1}{e}$$

$$\int_1^b \frac{1}{x} dx = 1$$

$$[\ln x]_1^b = 1$$

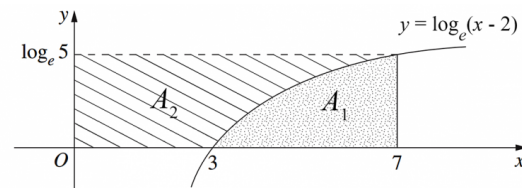
$$\ln b - \ln 1 = 1$$

$$\ln b = 1$$

$$\therefore b = e$$

IMPORTANT: Note when $\log_e a = 1$, the definition of a log means that $e^1 = a$. Many students failed to earn an easy 3rd mark by recognising this.

35. Calculus, 2ADV C4 2008 HSC 10a



$$\text{Shaded Area } (A_1) = \text{Rectangle} - A_2$$

$$\text{Area of Rectangle} = 7 \times \log_e 5$$

Finding the Area of A_2

$$y = \log_e(x-2)$$

$$x-2 = e^y$$

$$x = e^y + 2$$

$$\begin{aligned}
 \therefore A_2 &= \int_0^{\log_e 5} x dy \\
 &= \int_0^{\log_e 5} e^y + 2 dy \\
 &= [e^y + 2y]_0^{\log_e 5} \\
 &= [(e^{\log_e 5} + 2\log_e 5) - (e^0 + 0)] \\
 &= (5 + 2\log_e 5) - 1 \\
 &= 4 + 2\log_e 5
 \end{aligned}$$

$$\begin{aligned}
 \therefore A_1 &= 7\log_e 5 - (4 + 2\log_e 5) \\
 &= 5\log_e 5 - 4 \text{ u}^2
 \end{aligned}$$